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Split Stack Triple Height Cell

Abstract

Split stack triple height cells and methods of generating layouts of same are described herein. The structure includes a circuit formed within three stacked rows. The circuit includes a first stage having a first plurality of electrical components and a second stage having a second plurality of electrical components. The first row includes a first electrical component of the first plurality of electrical components within a top portion of the first row. A first electrical component of the second plurality of electrical components is within a bottom portion of the first row and a top portion of the second row. A second electrical component of the second plurality of electrical components is within a top portion of the third row and a bottom portion of the second row. A second electrical component of the first plurality of electrical components is within a bottom portion of the third row.

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Background/Summary

PRIORITY CLAIM [0001] This application is a continuation of U.S. application Ser. No. 18/348,430, filed Jul. 7, 2023, which is a continuation of U.S. application Ser. No. 17/148,652, filed Jan. 14, 2021, now U.S. Pat. No. 11,721,698, issued Aug. 8, 2023, which claims priority to U.S. Application No. 63/012,304, filed Apr. 20, 2020, the contents of each of which is incorporated by reference herein in their entirety.

FIELD

[0002] The technology described in this disclosure relates generally to device layouts and more particularly to split stack triple height cell.

BACKGROUND

[0003] Electronic Design Automation (EDA) and related tools enable efficient design of complex integrated circuits which may have extremely large numbers of components (e.g., thousands, millions, billions, or more). Specifying characteristics and placement of all of those components (e.g., transistor arrangements to implement desired logic, types of transistors, signal routing) by hand would be extremely time consuming and expensive for modern integrated circuits, if not impossible. Modern EDA tools utilize cells to facilitate circuit design at different levels of abstraction. A cell in the context of EDA is an abstract representation of a component within a physical device layout of an electronic circuit in software. Circuits may be designed using cells, where those circuits may then be implemented using lower level specifications (e.g., transistor arrangement, signal routing) associated with those cells. Standard libraries are used to design electronic circuits, enabling power-performance-area (PPA) optimization.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0005] FIG. 1A illustrates a block diagram of an exemplary electronic device implemented with staged circuits in accordance with various embodiments of the present disclosure.

[0006] FIG. 1B illustrates a layout arrangement of split stack triple height cell in accordance with various embodiments of the present disclosure.

[0007] FIG. 2 illustrates a two stage CMOS diagram in accordance with various embodiments of the present disclosure.

[0008] FIG. 3A illustrates an exemplary two-stage logic diagram of a buffer that can be implemented in a split stack triple height cell in accordance with various embodiments of the present disclosure.

[0009] FIG. 3B illustrates an exemplary two-stage transistor level schematic of the two-stage buffer

of FIG. 3A in accordance with various embodiments of the present disclosure.

[0010] FIG. 4 illustrates an example device layout of the buffer in FIG. 3B implemented using planar transistors within a split stack triple height cell in accordance with various embodiments of the present disclosure.

[0011] FIG. 5 illustrates another example device layout of the buffer in FIG. 3B implemented using FINFETs within a split stack triple height cell in accordance with various embodiments of the present disclosure.

[0012] FIG. 6A illustrates another exemplary two-stage transistor level schematic of an AND gate that can be implemented in a split stack triple height cell in accordance with various embodiments of the present disclosure.

[0013] FIG. 6B illustrates an example device layout of an AND gate of FIG. 6A implemented within a split stack triple height cell in accordance with various embodiments of the present disclosure.

[0014] FIG. 7 is an exemplary flow chart illustrating a method for generating a standard cell library having split stack triple height cells in accordance with various embodiments of the present disclosure.

[0015] FIG. 8 is an exemplary flow chart illustrating a computer-implemented method of generating a device layout of a split stack triple height cell in accordance with various embodiments of the present disclosure.

[0016] FIG. 9 is a block diagram depicting an electronic circuit design engine in accordance with various embodiments of the present disclosure.

[0017] FIG. 10 is a block diagram depicting modules of a circuit design engine in accordance with various embodiments of the present disclosure.

[0018] FIG. 11 is an exemplary block diagram illustrating a sample computing device architecture for implementing various aspects described herein in accordance with various embodiments of the present disclosure.

DETAILED DESCRIPTION

[0019] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0020] An Integrated Circuit (IC) is a complex network of a very large number of components (e.g., transistors, resistors, and capacitors) interconnected using the features of a process technology to realize a desired function. Manually designing such a component is typically not feasible because of the number of steps involved and the amount of design information that needs to be processed. EDA tools may be used to assist the designers in this process. Due to the size and complex nature of the design process, the IC may be designed using a hierarchical approach where the design is broken down in smaller pieces which are assembled to form the complete chip. This process also helps in pre-designing commonly used sub-blocks and reusing them where needed. A standard cell library is one such collection of basic components (e.g., AND, OR, NAND, NOR, XOR, flip-flops, latches) commonly used by certain EDA tools to automate the generation of layout from a behavioral description of a block. Each piece of design may have an abstract representation for the various information that is needed to capture the design such as functional

behavior, circuit description, physical layout, timing behavior, many of which are used by the EDA tools to assist in the design process.

[0021] EDA tools may use a library of standard cells associated with common circuit functions. For example, standard cells can be associated logic gates, such as an AND gate, an OR gate, an XOR gate, a NOT gate, a NAND gate, a NOR gate, or an XNOR gate, and circuits such as a multiplexer, a flip-flop, an adder, or a counter. Those standard cells can be arranged to realize more complex integrated circuit (IC) functions. When designing an IC having specific functions, standard cells may be selected. Designers, EDA software, and/or Electronic Computer-Aided Design (ECAD) tools can draw out design layouts of the integrated circuit including the selected standard cells and/or non-standard cells. The design layouts may be converted to photomasks. Then, semiconductor integrated circuits can be manufactured, when patterns of various rows, defined by the photomasks, are transferred to a substrate. More specifically, in addition to describing the functional operation of a cell, a standard cell may also include definition of components to realize that function as well as an arrangement of those components in a physical integrated circuit. Those components may be arranged in a single plane, or in some instances, the components may be stacked vertically over multiple rows to minimize integrated circuit area.

[0022] A digital IC consists of a large network of basic logical gates like AND, OR, NAND, NOR, EXOR, etc. A Standard cell library is a collection of these types of basic logical gates implemented in any given process technology. CAD tools are used to generate the IC layout from a behavioral description of the needed functionality. These CAD tools use the components from the standard cell library to realize the desired function. A cell in the standard cell library could be a basic logic gate implemented as a single stage CMOS circuit or multiple stages in which the output of one CMOS stage is connected to the input of the next stage to realize more complex function. Each CMOS stage consists of a network of P-type devices and a network of N-type devices connected together. The layout of each CMOS stage consists of a P-well region in which all the N-type devices are placed and an N-well region where all the P-type devices are placed. These devices are interconnected using the interconnection layers of the process technology to realize the circuit. To build a standard cell library, a convenient cell height is chosen and all the cells in the library are drawn with this same height or in some cases an integral multiple of the standard height. The width of each cell varies as needed to accommodate all of the devices and interconnect. The standard height of the library is divided into a top N-well portion where the P-type devices are placed and a bottom P-well portion where the N-type devices are placed. Typically, the P-well and N-well portions are equal. As mentioned earlier a standard cell could consist of a single stage or two stages and in some cases more than two CMOS stages. The widths of the N and P devices of a CMOS stage determines its drive strength, or its ability to drive a stable signal to the next cell that is connected to. These widths also determine the capacitive load that it presents to a cell that is driving it. The larger the device widths, the larger drive strength and also the larger its input capacitance. It is desirable to have a large drive strength but low input capacitance for any cell in the library. A standard cell library typically consists of multiple drive strength cells for any given logic function. A cell with two CMOS stages is typically built with the first stage using smaller devices so as to minimize the input pin capacitance and the second stage using larger devices in order to maximize its drive strength. This patent in some embodiments describes a novel layout scheme for such two stage cells where the devices that make up the 2 CMOS stages are efficiently placed across 3 vertically stacked standard cell heights.

[0023] IC functions may be performed in sequential stages, where outputs of a first stage are used as input to a second stage. In some cases, stage functionality can be stacked vertically in rows of a standard cell to maximize use of circuit area (e.g., a split stack triple height cell as described further herein). Systems and methods are provided herein that include staged circuits (e.g., having two or more stages) implemented to fit within three or more rows (e.g., split stack triple height cell). The staged circuits can be those implemented with planar transistor technology, FinFet process

technologies, and the like. More specifically, components of individual stages of a multi-stage circuit can be split across at least three rows of a device layout. The dividing of multi-stage IC circuitry across rows within cells and across separate cells may be strategically performed to realize design goals. For example, some embodiments may divide sub-circuitry or across different cells according to characteristics of the sub-circuitry such as physical sizes of components within the sub-circuitry.

[0024] FIG. 1A illustrates a block diagram of an exemplary electronic device **100** implemented with staged circuits in accordance with various embodiments of the present disclosure. Electronic device **100** is implemented in at least two stages (e.g., a first stage and a second stage). In this example, the depiction of circuit **110** of the first stage, and the physical components of its implementation, are smaller than circuit **120** of the second stage because the first stage is designed to receive and generate drive strength less than the second stage which handles greater drive strength. FIG. 1B illustrates a block diagram of split stack triple height cell **150** in accordance with various embodiments of the present disclosure. Electronic device **100** of FIG. 1A is implemented within what is referred to herein as a split stack triple height cell **150**, where components of individual stages of a multi-stage design are split among multiple rows of an at least three row tall structure. Split stack triple height cell **150** includes at least three rows **152**, **154**, **156** stacked on top of each other (e.g., triple height). The components of circuits **110**, **120** are implemented across the stacked rows **152**, **154**, **156**. More specifically, the components of circuit **110** can be split across rows **152**, **156**. The components of the larger circuit **120** can be split across multiple cells stacked within the structure **150**. For example, circuit **120** can be implemented within as row **154** as well as a portion of row **152** and row **156**. The components of smaller circuit **110** are split across a top portion of row **152** and a bottom portion of row **156**.

[0025] FIG. 2 illustrates a two stage CMOS function where in the output of the first stage **110** is connected to the input of the second stage **120** in accordance with various embodiments of the present disclosure. Each of the stages **110** and **120** includes a network of P devices (e.g., P stacks **202**, **204**) and a network of N devices (e.g., N stacks **206**, **208**). The P stack **202** and N stack **206** of the first stage **110** are sized smaller compared to the size of the P stack **204** and N stack **208** of the second stage **120** in order to minimize input pin capacitance and maximize drive strength. This relative sizing is represented in FIG. 2 by smaller boxes for the first stage **110** and larger boxes for the second stage **120**. FIG. 1B illustrates a generic split stack triple height cell layout arrangement for the 2 stage generic schematic in FIG. 1A in accordance with various embodiments of the present disclosure.

[0026] Turning back to FIG. 1B, each row **152**, **154**, **156** is made up of a P and N well region as described in more detail in FIG. 4. The middle row (e.g., cell **154**) is flipped with respect to the top row (e.g., cell **152**) and bottom row (e.g., cell **156**) such that the P-well region of the cell **152** abuts with the P-well region of the middle row (e.g., cell **154**). The N-well region of the third row (e.g., cell **156**) abuts with the N-well region of the middle row (e.g., cell **154**). This results in a pattern of smaller N-well region followed by larger P-well region followed by larger N-Well region followed by a smaller P-well region from top to bottom. Accordingly, the topmost N-well region of cell **152** and the bottommost P-well region of cell **154** accommodate smaller width P device and N device respectively and the middle P-well region and N-well regions (e.g., cells **152**, **154**, **156**) will accommodate a larger width N device and P device respectively. FIG. 1B also illustrates the placement of the various N and P devices in the CMOS schematic of FIG. 2 in this triple stack arrangement. The P stack **202** of the first stage **110** are placed in the topmost smaller N-well region and the N stack **206** of the first stage **110** are placed in the bottommost smaller P-well region. The N stack **208** of the second stage **120** are placed in the larger merged P-well region formed between the first and second rows (e.g., cells **152**, **154**) and the P stack **204** of the second stage **120** are placed in the larger merged N-well region formed between the second and third rows (e.g., cells **154**, **156**).

[0027] FIG. 3A illustrates an exemplary two-stage logic diagram **300** that can be implemented in a split stack triple height cell in accordance with various embodiments of the present disclosure. The two-stage logic diagram **300** depicted in FIG. 3A is a buffer having a first inverter **302** and a second inverter **304**. A buffer is an electronic circuit that outputs the same logic value as the input (e.g., if a logic low '0' is provided as input to a buffer, it will output a logic low '0'). Although the buffer outputs the same logic value that it receives as input, in the depicted example, the drive strength of the signal input to the buffer is amplified by the buffer. In other words, the output signal from the depicted buffer is of a greater drive strength than the input signal. The greater drive strength is used to drive a large circuit using the logic state of the original input signal.

[0028] The two-stage logic diagram **300** receives an input signal, I. The first inverter **302** outputs an inverted signal, A, to the second inverter **304**. The second inverter **304** is electrically coupled in series with the first inverter **302**. The second inverter **304** generates an output signal, O, that is substantially equivalent to the input signal, I, of the first inverter **302**. The first inverter **302** is visually illustrated as a smaller device than the second inverter **304**. The first inverter **302** is sized to be smaller than the second inverter **304** as it is designed to handle lower drive strength, and therefore requires smaller components. The second inverter **304** is visually illustrated as a larger device. The drive strength associated with the output signal, O, is larger than that of the input signal, I.

[0029] FIG. 3B illustrates an exemplary two-stage transistor level electrical schematic **350** of the two-stage logic diagram **300** of FIG. 3A in accordance with various embodiments of the present disclosure. The two-stage transistor level electrical schematic **350** includes two complementary metal-oxide-semiconductor field-effect transistor (CMOS) inverters **360**, **370** coupled together in series. CMOS inverter **360** is made up of a positive metal-oxide-semiconductor (PMOS) transistor **362** and a negative metal-oxide-semiconductor (NMOS) transistor **364**. CMOS inverter **370** is made up of a PMOS transistor **372** and a NMOS transistor **374**. CMOS inverter **370** can generate greater drive strength than the CMOS inverter **360** due to the physical sizing of each transistor. The CMOS inverters **360**, **370** can be implemented in a split stack triple height cell as illustrated in FIG. 4.

[0030] FIG. 4 illustrates an example device layout of the buffer in FIG. 3B implemented using planar transistors within a split stack triple height cell **400** in accordance with various embodiments of the present disclosure. For illustration purposes, the PMOS transistors **362**, **372** and NMOS transistors **364**, **374** are illustrated as planar transistors. Split stack triple height cell **400** includes three stacked rows **410**, **420**, **430**. A power rail and a ground rail are at the row boundaries bounding each cell. For example, row **410** is bounded by power rail **405** (e.g., VDD) and ground rail **415** (e.g., VSS). Row **420** is bounded by ground rail **415** (e.g., VSS) and power rail **425** (e.g., VDD). Row **430** is bounded by power rail **425** (e.g., VDD) and ground rail **435** (e.g., VSS). The power and ground rails provide power to the components implemented within the rows.

[0031] Each row has a P-well and an N-well. For example, first row **410** includes P-well **414** and N-well **416**. Second row **420** includes P-well **424** and N-well **426**. Third row **430** includes P-well **434** and N-well **436**. P-well **414** and P-well **424** abut each other and create a larger P-well area (e.g., approximately double in size) than the area covered by P-well **434**. Similarly, N-well **426** and N-well **436** abut each other and create a larger N-well area (e.g., approximately double in size) than the area covered by N-well **416**. The merged P-well region (e.g., P-well **414** and P-well **424**) and the merged N-well region (e.g., N-well **426** and N-well **436**) each facilitate diffusion across the area between rows **410**, **420**, **430**. Use of the area between the rows **410**, **420**, **430** facilitates use of more area of the triple stack cell height structure **400**. In other words, less area of the triple stack cell height structure goes unused. The CMOS inverters **360**, **370** are implemented within the rows **410**, **420**, **430**. More specifically, PMOS transistors are implemented within N-wells and NMOS transistors are implemented in P-wells. Because CMOS inverter **370** is larger in size than CMOS inverter **360**, the components of CMOS inverter **370** are implemented across the larger P-well and

N-well areas. In other words, NMOS transistor **374** is implemented within the larger P-well area formed by P-well **414** and P-well **424**. PMOS transistor **372** is implemented within the larger N-well area formed by N-well **426** and N-well **436**. A polysilicon interconnect **440** connects the PMOS transistor **372** with NMOS transistor **374** to form the CMOS inverter **370**. Because CMOS inverter **360** is smaller in size than CMOS inverter **370**, CMOS inverter **360** can be split across row **410** and row **430**. PMOS transistor **362** is implemented in N-well **416**. NMOS transistor **364** is implemented in P-well **434**.

[0032] FIG. 5 illustrates another example layout of the buffer in FIG. 3B implemented using FINFETs within a split stack triple height cell **500** in accordance with various embodiments of the present disclosure. For illustration purposes, the PMOS transistors **362**, **372** and NMOS transistors **364**, **374** are illustrated as FinFet transistors. Split stack triple height cell **500** includes rows **510**, **520**, **530**. In the embodiment illustrated in FIG. 5, CMOS inverter **370** is implemented in rows **510**, **520**, **530**. More specifically, PMOS transistor **372** having five P-fins **538** including P-fin **542** is implemented in the larger N-well area formed by the combination of N-well **526** of row **520** and N-well **536** of row **530**. P-fin **542** in some embodiments may be a dummy fin. In other embodiments, P-fin **542** may be an active fin. NMOS transistor **374** having five N-fins **528** is implemented in the larger P-well area formed by P-well **514** of row **510** and P-well **524** of row **520**. N-fin **540** in some embodiments may be a dummy fin. In other embodiments, N-fin **540** may be an active fin. Polysilicon **550** forms the gate for the NMOS transistor **374** and PMOS transistor **372** to form the CMOS inverter **370**. The polysilicon **550** also connects the gates of NMOS transistor **374** and PMOS transistor **372** together as illustrated in FIG. 3B.

[0033] CMOS inverter **360** is split across a top portion of row **510** and a bottom portion of row **530**. More specifically, PMOS transistor **362** having two P-fins **518** is implemented within N-well **516** of row **510** (e.g., top portion of row **510**). NMOS transistor **364** having two N-fins **534** is implemented within P-well **537** of row **530** (e.g., bottom portion of row **530**). The layout is completed along with the power rails and other connections between the devices in accordance with the circuit schematic of FIG. 3B using the interconnect layers.

[0034] FIG. 6A illustrates another exemplary two-stage transistor level schematic of an AND gate **600** that can be implemented in a split stack triple height cell in accordance with various embodiments of the present disclosure. An AND gate is a logic gate that outputs a logic high when both inputs to the AND gate are logic high and a logic low when either of the inputs is at a logical low level. The AND gate **600** includes two stages: a NAND gate **610** and inverter **620**. The NAND gate **610** includes PMOS transistors **612**, **614** and NMOS transistors **616**, **618**. A NAND gate is a logic gate that outputs a logic low when both inputs are high and a logic high when either of the inputs are at logic low level. Inverter **620** includes PMOS transistor **622** and NMOS transistor **624**. Inverter **620** is constructed using larger devices than that of NAND gate **610**.

[0035] FIG. 6B illustrates an example layout of the AND gate **600** of FIG. 6A implemented within a split stack triple height cell **650** in accordance with various embodiments of the present disclosure. Split stack triple height cell **650** includes three stacked rows **660**, **670**, **680**. For illustration purposes, the PMOS transistors **612**, **614**, **622** and NMOS transistors **616**, **618**, **624** are illustrated as FinFet transistors. Row **660** includes P-well **662** and N-well **663**. Row **670** includes P-well **672** and N-well **673**. Row **680** includes P-well **682** and N-well **683**. Because NAND gate **610** uses transistors with lesser fins than that of inverter **620**, NAND gate **610** can be split across a top portion of row **660** and a bottom portion of row **680**. More specifically, PMOS transistor **612** having two P-fins **664** and PMOS transistor **614** having two P-fins **665** are both implemented in N-well **663**. NMOS transistor **616** having N-fins **684** and NMOS transistor **618** having N-fins **686** are both implemented in P-well **682**. Because inverter **620** uses larger number of fins than NAND gate **610**, PMOS transistor **622** and NMOS transistor **624** are implemented within larger N-well and P-well areas. More specifically, NMOS transistor **624** having fins **674** is implemented across two legs or columns **692**, **694** within the larger P-well created by abutting P-well **662** and P-well **672**. N-fin

676 in some embodiments may be a dummy fin. In other embodiments, N-fin **676** may be an active fin. PMOS transistor **622** having P-fins **687** is implemented across two legs or columns **692**, **694** within the larger N-well area created by abutting N-well **673** and N-well **683**. P-fin **688** in some embodiments may be a dummy fin. In other embodiments, P-fin **688** may be an active fin.

[0036] The merged P-well region (e.g., P-well **662** and P-well **672**) and the merged N-well region (e.g., N-well **673** and N-well **683**) each facilitate active diffusion along the intersection area between rows **660**, **670**, **680**. Use of the intersection area between the rows **660**, **670**, **680** for active diffusion results in increased diffusion width per leg in the triple height structure. This in turn results in increased drive strength of the cell for the same cell area. Use of more area within the triple stack cell height structure **600** also enables a lesser number of legs within the structure to achieve a given drive strength. With a lesser number of legs, the wiring needed to connect different legs is reduced, in turn reducing the interconnect parasitics. With lesser parasitics, the performance and power of the cell improves. This also results in lesser routing blockage within the cell, improving signal routability. The pins are also distributed over three rows **660**, **670**, **680** further increases pin accessibility for routing.

[0037] FIG. 7 is an exemplary flow chart **700** illustrating a method to determine if each cell in the standard cell library is a good candidate for implementing as a split stack triple height cells in accordance with various embodiments of the present disclosure. While FIG. 7 is described here with reference to previously described structures for ease in understanding, it is understood that the method applies to many other structures as well. In order to generate a split stack cell height structure, such as those illustrated in FIGS. 3B, 4, and 6B, a maximum number of P or N Fins that can be implemented per leg within a single standard cell height is determined (e.g., step **702**). For example, considering row **660** the maximum number of P fins or N fins that can be implemented in one leg or column, **692** within this row is two (e.g., two P-fins **664** and two N-fins **674**). Similarly the maximum number of fins for row **660** within the leg or column **694** is also two (e.g., two P-fins **665** and two N-fins **674**). A maximum fin count per leg for merged diffusion is then determined (e.g., step **704**). This fin count will include the two fins per leg within each row and an additional fin at the intersection of the rows. This fin count accounts for the fins between similar wells of abutting rows. For example, P-well **662** of row **660** abuts P-well **672** of row **670** each of which can accommodate two fins. In addition, a fin **676** can be placed at the intersection of the rows **660**, **670**. In this example, the maximum fin count for leg or column **694** is five. The cell is first implemented as a standard single height cell and its PPA is evaluated (e.g. step **706**). Next it is determined if the cell is a stage CMOS gate (e.g., step **708**). If the cell is a single stage CMOS gate, the cell is not considered for split stack triple height implementation and analysis is shifted to the next cell in the standard cell library (e.g. step **726**) and the process is repeated starting back at step **706**. If the cell consists of two CMOS stages, then P-fins of the first stage (e.g., P-fins **518** of PMOS transistor **362** or P-fins **664** of PMOS transistor **612** and P-fins **665** of PMOS transistor **614**) are implemented within a top portion of the first row (e.g., N-well **516** of row **510** or N-well **663** of row **660**) of the triple stacked cell height structure (e.g., step **710**). The PMOS device may need to be implemented with multiple legs if the number of P fins needed exceed the maximum number of fins per leg within one row. Next, N-fins of the first stage device (e.g., N-fins **534** of NMOS transistor **364** or N-fins **684** of NMOS transistor **616** and N-fins **686** of NMOS transistor **618**) are implemented in the bottom half of the third row (e.g., P-well **537** of row **530** or P-well **682** of row **680**) (e.g., step **712**). The N-fins of the second stage device (e.g., N-fins **528** of NMOS transistor **374** or N-fins **674** of NMOS transistor **624**) are implemented within the larger P-well area of abutting P-wells (e.g., P-well **514** and P-well **524** or P-well **662** and P-well **672**) between the first and second rows (e.g., row **510** and row **520** or row **660** and row **670**) based on the maximum number of fins per leg for the merged diffusion (e.g., step **714**). Using the example triple stack cell height structure **650**, the number of N-fins **674** implemented per leg (e.g., leg **692** or leg **694**) is five (e.g., two N-fins in P-well **662**, two N-fins in P-well **672** and one N-fin **676** between P-well **662** and P-well **672**). The P-

pins of the second stage device (e.g., P-fins **538** of PMOS transistor **372** or P-fins **687** of PMOS transistor **622**) are implemented within the larger N-well area of abutting N-wells (e.g., N-well **526** and N-well **536** or N-well **673** and N-well **683**) between the second and third rows (e.g., row **520** and row **530** or row **670** and row **680**) based on the maximum number of fins per leg for the merged diffusion (e.g., step **716**). Once the various transistors and fins are placed within the layout, the layout is completed by establishing all the interconnects as per the circuit schematic (e.g., step **718**). PPA analysis is performed through simulation using the generated device layout (e.g., step **720**). It is then determined whether the PPA analysis is better than that output during step **706** (e.g., step **722**). If the PPA analysis is better than that output during step **706**, then the cell is added to a cell library such as those described by FIGS. **8-9** (e.g., step **724**) and the next cell is selected analysis (e.g., step **726**). If the PPA analysis, on the other hand, is not better than that output during step **706**, then the cell is not added to the library and the next cell is selected from the cell library (e.g., step **726**).

[0038] FIG. **8** is an exemplary flow chart **800** illustrating a computer-implemented method of generating a device layout of a split stack triple height cell in accordance with various embodiments of the present disclosure. While FIG. **8** is described here with reference to previously described structures for ease in understanding, it is understood that the method applies to many other structures as well. A computer-implemented method of defining a circuit layout (e.g., split stack triple height cell **400**, split stack triple height cell **500**, split stack triple height cell **650**) including a plurality of stacked rows (e.g., rows **410**, **420**, **430**; rows **510**, **520**, **530**; rows **660**, **670**, **680**) includes a number of steps. A cell library such as cell repository **910** includes split stack triple height cell (e.g., split stack triple height cell **400**, split stack triple height cell **500**, split stack triple height cell **650**). A first row (e.g., row **410**; row **510**; row **660**) including a first P-well (e.g., **414**; **514**; **662**) and a first N-well (e.g., **416**; **516**; **663**) is defined within the cell library such as cell repository **910** (e.g., at step **810**). A P-device (e.g., PMOS transistor **362**; PMOS transistor **612** or PMOS transistor **614**) of a first electrical component (e.g., CMOS inverter **360**; NAND gate **610**) is formed in the first N-well (e.g., N-well **416**; N-well **516**; N-well **663**). A second row (e.g., row **420**; row **520**; row **670**) is defined within the cell library (e.g., cell repository **910**) beneath the first row such as row **410**, row **510**, or row **660** (e.g., step **820**). The second row (e.g., row **420**; row **520**; row **670**) includes a second P-well (e.g., P-well **424**; P-well **524**; P-well **672**) and a second N-well (e.g., N-well **426**; N-well **526**; N-well **673**). An N-device (e.g., NMOS transistor **374**; NMOS transistor **624**) of a second electrical component (e.g., CMOS inverter **370**; inverter **620**) is formed within the first P-well (e.g., P-well **414**; P-well **514**; P-well **662**) and the second P-well (e.g., P-well **424**; P-well **524**; P-well **672**). A third row (e.g., row **430**; row **530**; row **680**) beneath the second row (e.g., row **420**; row **520**; row **670**) is defined within the cell library such as cell repository **910** (e.g., step **830**). The third row (e.g., row **430**; row **530**; row **680**) includes a third P-well (e.g., P-well **434**; P-well **537**; P-well **682**) and a third N-well (e.g., N-well **436**; N-well **536**; N-well **683**). A P-device (e.g., PMOS transistor **372**; PMOS transistor **622**) of the second electrical component (e.g., CMOS inverter **370**; inverter **620**) is formed within the second N-well (e.g., N-well **426**; N-well **526**; N-well **673**) and the third N-well (e.g., N-well **436**; N-well **536**; N-well **683**) and an N-device (e.g., NMOS transistor **364**; NMOS transistor **616** or NMOS transistor **618**) of the first electrical component (e.g., CMOS inverter **360**; NAND gate **610**) is formed within the third P-well (e.g., P-well **434**; P-well **537**; P-well **682**). The circuit layout such as split stack triple height cell **400**, split stack triple height cell **500**, or split stack triple height cell **650**, includes the first row (e.g., row **410**; row **510**; row **660**), the second row (e.g., row **420**; row **520**; row **670**), and the third row (e.g., row **430**; row **530**; row **680**) stacked on top of each other.

[0039] FIG. **9** is a block diagram depicting an electronic circuit design engine in accordance with various embodiments of the present disclosure. The electronic circuit design engine **902** facilitates development of a production integrated circuit design **904** that is used in the fabrication of a physical integrated circuit. The electronic circuit design engine **902** receives or facilitates initial

generation of an integrated circuit design **906** that may be developed (e.g., over a number of iterative revisions) and stored in a non-transitory circuit design repository **908**, such as via interactions with a user interface or execution of automated scripts. For example, on request, the electronic circuit design engine **902** may access or receive the integrated circuit design **906** in the form of a computer file, perform operations on the integrated circuit design **906**, and then output a modified form of the design (e.g., as an integrated circuit design **906** file for storage in the design repository **908** or as a production integrated circuit design **904** (e.g., in the form of an EDA file, a netlist) for fabrication). The integrated circuit design **906** may be made up of a plurality of components (e.g., resistors, capacitors, transistors logic gates, data signal lines), some or all of which take the form of cells. The integrated circuit design **906** may take a variety of forms, such as a behavioral model of a design in a register-transfer level (RTL) representation or a more hardware specific specification, such as a netlist. The electronic circuit design engine **902** is responsive to one or more cell repositories (e.g., cell repository **910**) that store data associated with cells that can be used as building blocks in the generation of integrated circuit designs **904**, **906**. Such cells can include standard cells that may take a variety of forms and represent a variety of functions (e.g., the operation of one or more logic gates), such as split triple stack cell height structures implementing two or more stages of a logic gate.

[0040] Electronic circuit design engines may provide a variety of different circuit design functionality. FIG. **10** is a block diagram depicting modules of a circuit design engine in accordance with various embodiments of the present disclosure. An electronic circuit design engine **902** receives an integrated circuit design **906** via a file or commands that dictate the content of that design **906** entered via a mechanism such as a circuit design user interface **1002**. The interface **1002** may display graphics or text describing an integrated circuit design and provide commands for building and manipulating the design. The circuit design user interface **1002** can be used, for example, place the cells within an integrated circuit design. For example, the circuit analysis and modification engine **1004** can automatically or semi automatically recommend optimized cell arrangements such as those described in FIGS. **1A-6B**. The electronic circuit design engine **902** is further responsive to a cell repository **910** that stores cell data records like the one depicted at **912** with varying heights which may not be a multiple of a standard single cell row height and with the placement reference edge at the intersection of the N-well and P-well. The circuit design user interface **1002** can provide controls for accessing standard cells from the repository **910** and integrating them into an integrated circuit design **906**. Upon completion of an integrated circuit design **906**, the design may be output from the electronic circuit design engine **902** for saving in a non-transitory computer readable medium or as a production integrated circuit design **904** for fabrication of an integrated circuit. Additionally, a circuit synthesizer **1006** can automatically compile and optimize the integrated circuit design **906**.

[0041] FIG. **11** is an exemplary block diagram **1100** illustrating a sample computing device architecture for implementing various aspects described herein. A bus **1104** can serve as the information highway interconnecting the other illustrated components of the hardware. A processing system **1108** labeled CPU (central processing unit) (e.g., one or more computer processors/data processors at a given computer or at multiple computers), can perform calculations and logic operations required to execute a program. A non-transitory processor-readable storage medium, such as read only memory (ROM) **1112** and random access memory (RAM) **1116**, can be in communication with the processing system **1108** and can include one or more programming instructions for the operations specified here. Optionally, program instructions can be stored on a non-transitory computer-readable storage medium such as a magnetic disk, optical disk, recordable memory device, flash memory, or other physical storage medium.

[0042] In one example, a disk controller **1148** can interface one or more optional disk drives to the system bus **1104**. These disk drives can be external or internal CD-ROM, CD-R, CD-RW or DVD, or solid state drives such as **1152**, or external or internal hard drives **1156**. As indicated previously,

these various disk drives **1152**, **1156** and disk controllers are optional devices. The system bus **1104** can also include at least one communication port **1120** to allow for communication with external devices either physically connected to the computing system or available externally through a wired or wireless network. In some cases, the communication port **1120** includes or otherwise comprises a network interface.

[0043] To provide for interaction with a user, the subject matter described herein can be implemented on a computing device having a display device **1140** (e.g., a CRT (cathode ray tube) or LCD (liquid crystal display) monitor) for displaying information obtained from the bus **1104** to the user and an input device **1132** such as keyboard **1136** and/or a pointing device (e.g., a mouse or a trackball) and/or a touchscreen by which the user can provide input to the computer. Other kinds of input devices **1132** can be used to provide for interaction with a user as well; for example, feedback provided to the user can be any form of sensory feedback (e.g., visual feedback, auditory feedback by way of a microphone or tactile feedback); and input from the user can be received in any form, including acoustic, speech, or tactile input. In the input device **1132** and the keyboard **1136** can be coupled to and convey information via the bus **1104** by way of an input device interface **1128**. Other computing devices, such as dedicated servers, can omit one or more of the display **1140** and display interface **1114**, the input device **1132**, the keyboard **1136**, and input device interface **1128**.

[0044] Additionally, the methods and systems described herein may be implemented on many different types of processing devices by program code comprising program instructions that are executable by the device processing subsystem. The software program instructions may include source code, object code, machine code, or any other stored data that is operable to cause a processing system to perform the methods and operations described herein and may be provided in any suitable language such as C, C++, JAVA, Perl, Python, Tels, for example, or any other suitable programming language. Other implementations may also be used, however, such as firmware or even appropriately designed hardware configured to carry out the methods and systems described herein.

[0045] The systems' and methods' data (e.g., associations, mappings, data input, data output, intermediate data results, final data results, etc.) may be stored and implemented in one or more different types of computer-implemented data stores, such as different types of storage devices and programming constructs (e.g., RAM, ROM, Flash memory, flat files, databases, programming data structures, programming variables, IF-THEN (or similar type) statement constructs, etc.). It is noted that data structures describe formats for use in organizing and storing data in databases, programs, memory, or other computer-readable media for use by a computer program.

[0046] The computer components, software modules, functions, data stores and data structures described herein may be connected directly or indirectly to each other in order to allow the flow of data needed for their operations. It is also noted that a module or processor includes but is not limited to a unit of code that performs a software operation, and can be implemented for example as a subroutine unit of code, or as a software function unit of code, or as an object (as in an object-oriented paradigm), or as an applet, or in a computer script language, or as another type of computer code. The software components and/or functionality may be located on a single computer or distributed across multiple computers depending upon the situation at hand.

[0047] Use of the various processes as described herein can provide a number of advantages. For example, use of the subject matter enables improved performance of stages electronic devices throughout improved power and signal routability. Additionally, splitting of the device transistors across multiple rows of a split stack triple height cell based on sizing of the staged components occupies a smaller surface area.

[0048] In one embodiment, a split stack triple height cell includes a circuit, a first row, a second row, and a third row. The circuit includes a first stage having a first plurality of electrical components and a second stage having a second plurality of electrical components. The first row

includes a first electrical component of the first plurality of electrical components within a top portion of the first row. The second row is stacked beneath the first row. A first electrical component of the second plurality of electrical components is within a bottom portion of the first row and a top portion of the second row and across a boundary between the first row and the second row. The third row is stacked beneath the second row. A second electrical component of the second plurality of electrical components is within a top portion of the third row and a bottom portion of the second row and across a boundary between the second row and the third row and wherein a second electrical component of the first plurality of electrical components is within a bottom portion of the third row.

[0049] In another embodiment, a device includes a first logic gate and a second logic gate. The first logic gate is formed within a first section of a multiple row structure and a fourth section of the multiple row structure. The second logic gate is formed within a second section of the multiple row structure and a third section of the multiple row structure. The first section and a portion of the second section forms a first row. Another portion of the second section and half of the third section forms a second row. Another portion of the third section and the fourth section forms a third row. The first row, the second row, and the third row are stacked on top of each other form the multiple row structure.

[0050] In yet another embodiment, a computer-implemented method of defining a device layout including a split stack triple height cell having a plurality of stacked rows. A first row including a first P-well and a first N-well is defined within a cell library (e.g., step **810**). A P-device of a first electrical component is formed in the first N-well. A second row beneath the first row is defined within the cell library (e.g., step **820**). The second row includes a second P-well and a second N-well. The N-device of a second electrical component is formed within the first P-well and the second P-well. A third row beneath the second row is defined within the cell library (e.g., step **830**). The third row includes a third P-well and a third N-well. The P-device of the second electrical component is formed within the second N-well and the third N-well and an N-device of the first electrical component is formed within the third P-well. Using an electronic placement tool the circuit layout is generated having the first row, the second row, and the third row stacked on top of each other (e.g., step **840**).

[0051] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A structure comprising: a first stage having a first electrical component; and a second stage having a second electrical component, wherein: an output of the first stage is an input of the second stage; the first electrical component is in a top portion of a first row; and the second electrical component is in a bottom portion of the first row and a top portion of a second row and across a boundary between the first and second rows.
2. The structure of claim 1, wherein the first stage is of a smaller physical size than the second stage.
3. The structure of claim 1, wherein: the first stage has a third electrical component; the second stage has a fourth electrical component; the fourth electrical component in a top portion of a third row and a bottom portion of the second row and across a boundary between the second and third

rows; and the third electrical component is in a bottom portion of the third row.

4. The structure of claim 3, further comprising a polysilicon interconnect across the bottom portion of the first row, the top and bottom portions of the second row, and the top portion of the third row, wherein the polysilicon interconnect couples together the second and fourth electrical components of the second stage.

5. The structure of claim 1, wherein each of the first and second stages comprises at least one of an inverter, an AND gate, a NAND gate, and an OR gate.

6. The structure of claim 3, further comprising a gate extending across the bottom portion of the first row, the top and bottom portions of the second row, and the top portion of the third row.

7. The structure of claim 1, wherein the first, second, and third rows each have at least one fin.

8. The structure of claim 3, further comprising: a first power rail and a first ground rail at a top outer boundary and a bottom outer boundary of the first row, respectively, wherein the first ground rail is at the boundary between the first and second rows; and a second power rail and a second ground rail at a top outer boundary and a bottom outer boundary of the third row, respectively, wherein the second power rail is at the boundary between the second and third rows.

9. The structure of claim 1, wherein the first electrical component or the second electrical component comprises FinFet transistors having a plurality of fins and wherein a number of fins of the first stage is less than a number of fins of the second stage.

10. The structure of claim 3, further comprising: a fifth electrical component of the first stage in the first row, wherein the first electrical component of the first stage is oriented in a first leg of the first row and the fifth electrical component of the first stage is oriented in a second leg of the first row; and a sixth electrical component of the first stage in the third row, wherein the third electrical component of the first stage is oriented in a first leg of the third row and the sixth electrical component of the first stage is oriented in a second leg of the third row.

11. The structure of claim 1, wherein the second electrical component of the second stage is split across first and second legs.

12. The structure of claim 11, wherein an active fin of each of the first and second legs is on the boundary between the first and second rows, aligning with a ground rail.

13. The structure of claim 1, wherein the fourth electrical component of the second stage is split across first and second legs.

14. The structure of claim 13, wherein an active fin of each of the first and second legs is on the boundary between the second and third rows, aligning with a power rail.

15. A device comprising: a first logic gate formed in first and fourth sections; and a second logic gate formed in second and third sections and including a gate region across the second and third sections, wherein: the first section and a first portion of the second section form a first row, a second portion of the second section and a first portion of the third section form a second row.

16. The device of claim 15, wherein: a second portion of the third section and the fourth section form a third row, the first, second, and third rows form a multiple row structure, and the second section is larger than the size of the first section and the third section is double the size of the fourth section.

17. The device of claim 15, wherein the first logic gate is of a smaller physical size than the second logic gate.

18. The device of claim 15, wherein the first or second logic gate comprises FinFet transistors.

19. A computer-implemented method of defining a circuit layout comprising a plurality of stacked rows, the computer-implemented method comprising: defining, within a cell library, a first row and a second row, wherein at least one of the first and second rows comprises a P-well and a N-well; defining, within the cell library, a power rail or a ground rail at an outer boundary of the first row, or a boundary between the first and second rows; and generating, the circuit layout comprising the first and second rows.

20. The computer-implemented method of claim 19, further comprising: defining, within a cell

library, a third row; and defining a circuit including first and second stages in the first, second, and third rows, wherein an output of the first stage is an input of the second stage, wherein each of the first, second, and third rows comprises a P-well and a N-well.
